



YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAM
Higher 2

NAME

INDEX NO

CG

BIOLOGY

9744/02

Paper 2 Structured Questions

31 Aug 2022

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, index no. and CG on this cover page.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer all questions in the spaces provided on the Question paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question

At the end of the examination, **submit booklets A, B and C separately** to the invigilator.

For Examiner's Use	
Section A	
1	12
2	9
3	10
4	7
5	9
6	9
7	9
8	11
9	12
10	7
11	5
Total	100

This document consists of **29** printed pages and **1** blank page.

Answer **all** questions

- 1 Collagen is a major component of the cartilage found in some of the structures of the human gas exchange system. Cells that synthesise and secrete the components of cartilage are known as chondrocytes.

Fig. 1.1 is a transmission electron micrograph of a chondrocyte.

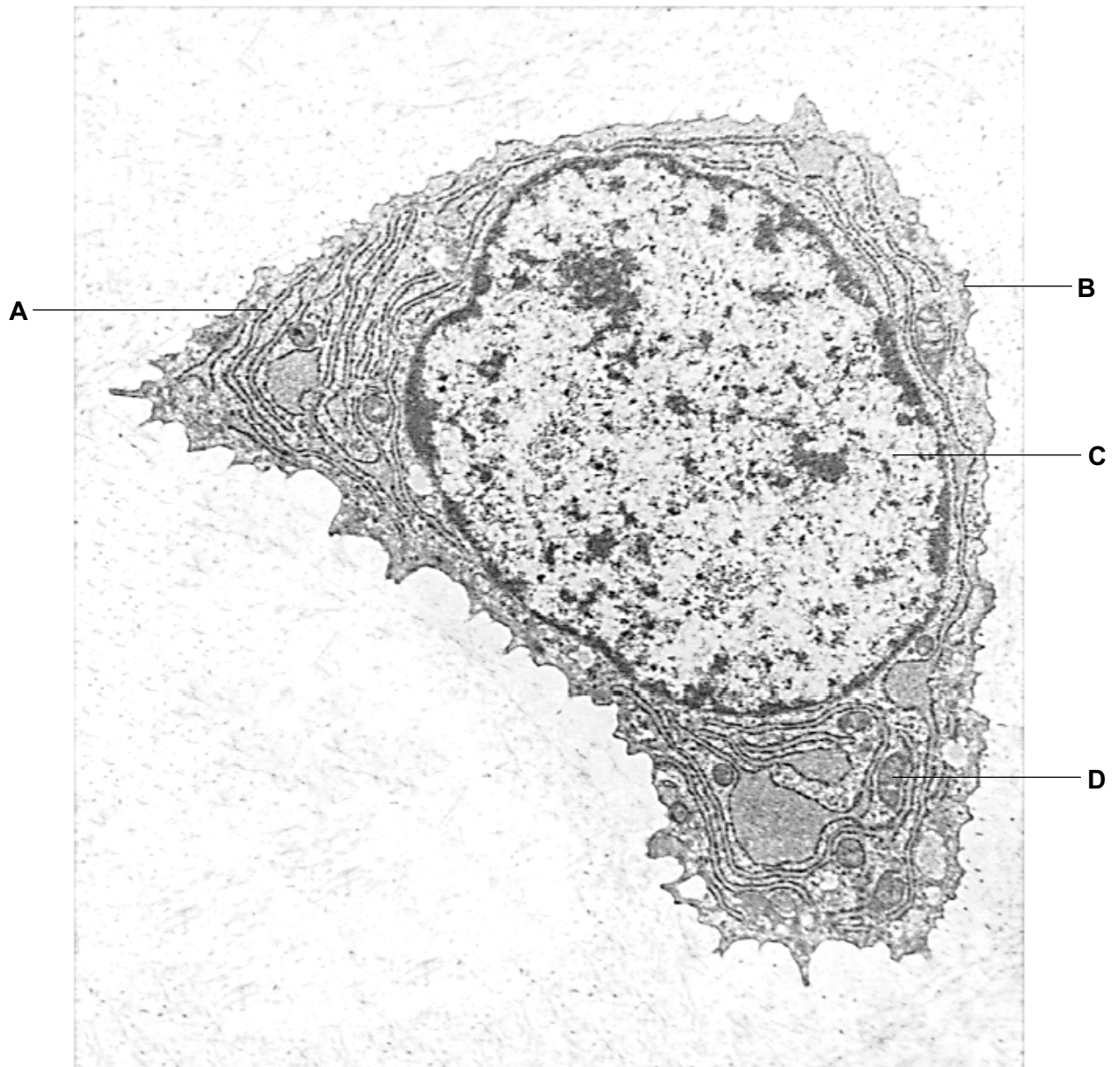


Fig. 1.1

(a) Identify structures **A - D**.

A

.....

B

.....

C

.....

D

.....

[2]

- (b) With reference to Fig. 1.1, explain **two** features of the chondrocyte that show how the cell is adapted to its function.

.....

.....

.....

.....

[2]

- (c) Collagen forms tropocollagen which is transported to the Golgi apparatus before being secreted out of the chondrocyte.

Describe the role of Golgi apparatus in collagen formation.

.....

.....

.....

[2]

- (d) Fig. 1.2 shows part of the primary structure of a collagen polypeptide.

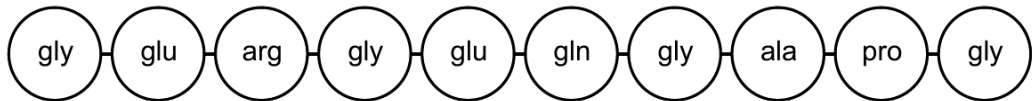


Fig. 1.2

With reference to Fig. 1.2, name the

- (i) type of covalent bond formed between the amino acids;

..... [1]

- (ii) enzyme that catalyses the formation of bond between the amino acids;

..... [1]

- (iii) reaction that forms the covalent bond between the amino acids.

..... [1]

(iv) Fig. 1.3 shows the molecular structure of the amino acid glycine (gly).

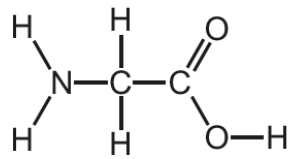


Fig. 1.3

With reference to Fig. 1.2 and Fig. 1.3 and the function of collagen, explain how the structure of a collagen polypeptide makes it suitable to form a collagen molecule.

.....

.....

.....

.....

.....

.....

.....

[3]

[Total: 12]

- 2 (a)** The content of lysosomes in animal cells has a pH of 5.0. The cytosol has a pH of 7.2. The lysosomal membrane enclosing the lysosomal content, controls the passage of hydrogen ions from the cytosol into the lysosome. The low pH created by the entry of hydrogen ions is optimum for the action of acid hydrolase enzymes in the lysosome.

- (i)** State and explain the transport mechanism used to move hydrogen ions from the cytosol of an animal cell into the lysosome.

Transport mechanism

Explanation

[3]

- (ii)** Suggest how the structure of the lysosomal membrane allows hydrogen ions to be transported into the lysosome but does not allow the ions to leave the lysosome.

[2]

- (iii)** The acid hydrolases in the lysosome cannot function in neutral conditions (pH 7.0) or alkaline conditions.

Explain the advantage to the animal cell of having acid hydrolases that cannot function in neutral, near neutral or alkaline conditions.

[2]

- (b) An analysis of a membrane protein located on the lysosomal membrane is carried out. The amino acid sequence that makes up the transmembrane segment of the protein is shown below.

...Ile - Thr - Leu - Ile - Tyr - Phe - Gly - Val - Met - Ala - Gly - Val - Ile - Gly - Thr - Ile - Leu - Leu - Ile - Ser...

Suggest why such an amino acid sequence would enable the protein to span the lysosomal membrane.

.....

.....

.....

.....

.....

[2]

[Total: 9]

- 3 Cyclooxygenase (COX) is an enzyme that catalyses the conversion of arachidonic acid (AA), into prostaglandins which are then involved in promoting inflammation, pain and fever.

There are two isoforms of the enzyme, COX-1 and COX-2 and they differ in the shape of their active sites.

Fig. 3.1 shows a simplified diagram of the active sites in COX-1 and COX-2.

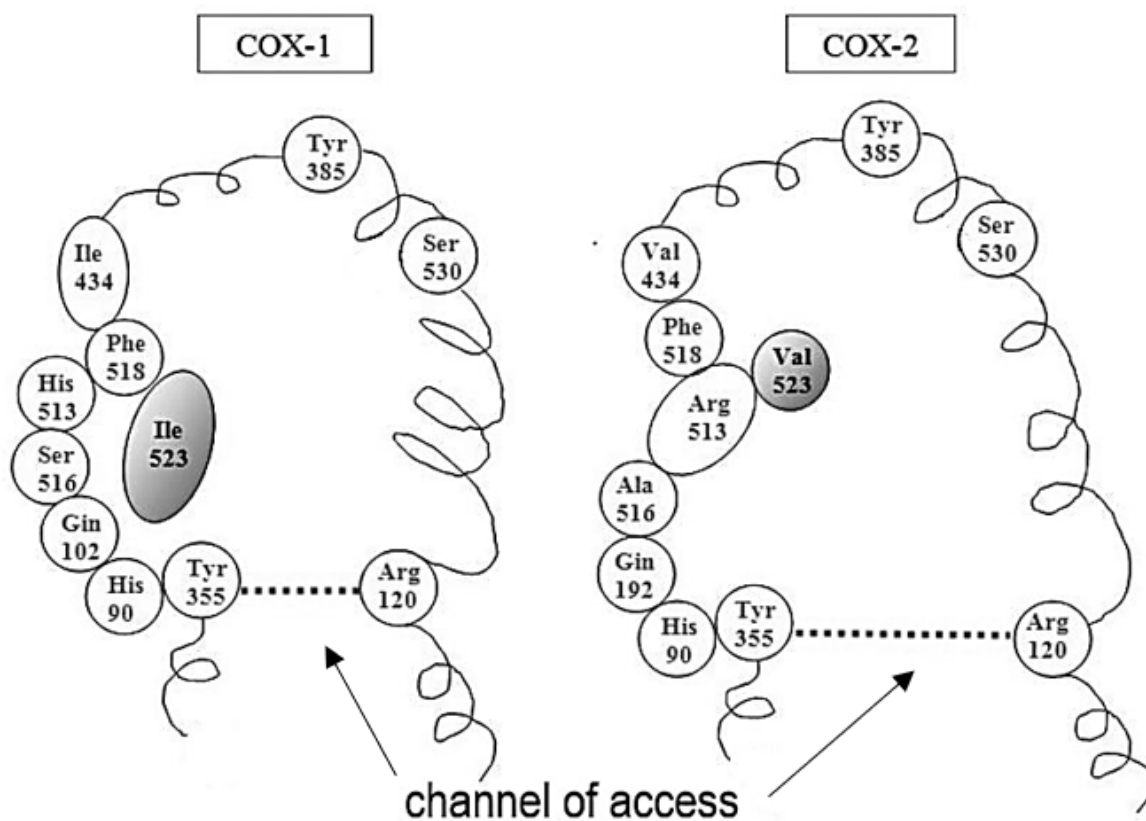


Fig. 3.1

- (a) With reference to Fig. 3.1, describe how the differences in amino acids result in a change in the configuration of the active site in COX-1 and COX-2.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[4]

COX enzymes are targets of nonsteroidal anti-inflammatory drugs (NSAIDs) e.g., Ibuprofen, to reduce the inflammatory symptoms.

Fig. 3.2 shows COX-1 when bound to AA and when ibuprofen is present.

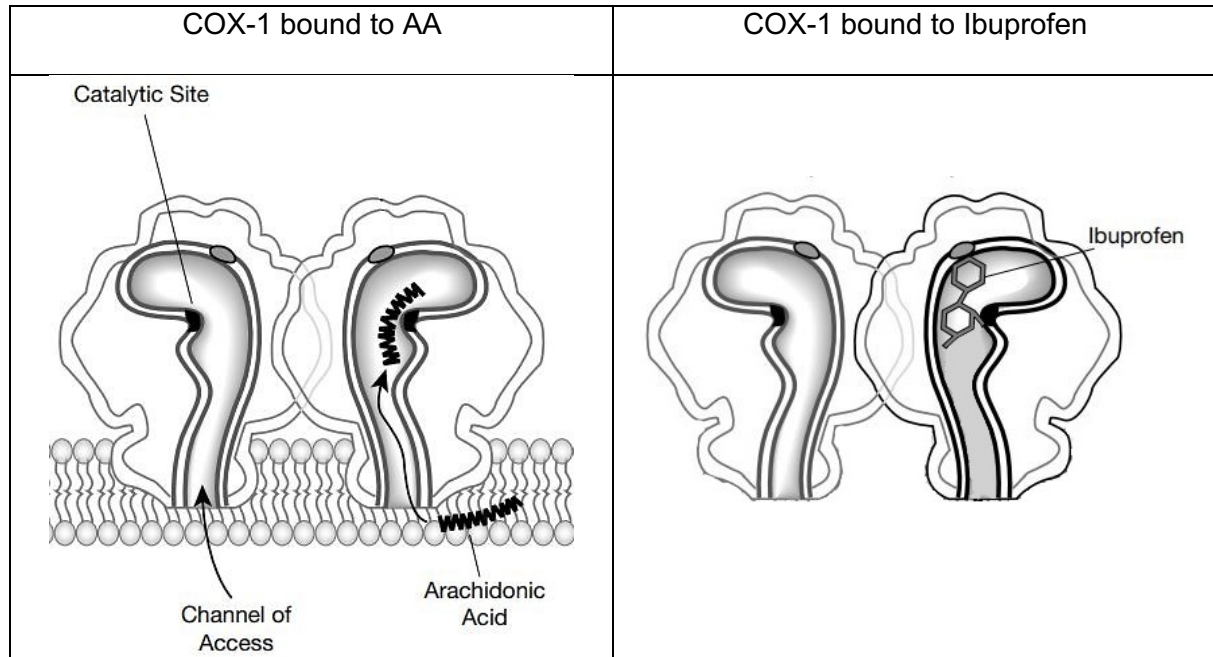


Fig. 3.2

(b) With reference to Fig. 3.2,

(i) explain how ibuprofen reduces the effects of inflammation.

.....

.....

.....

.....

.....

.....

.....

[3]

(ii) explain how ibuprofen can bind to the same active site.

.....

.....

.....

.....

[2]

- (c) Inhibition of COX-1 by NSAIDs has been strongly implicated in gastric ulceration and bleeding. Newer drugs have been developed to selectively inhibit only COX-2.

Using information from Fig. 3.1 and 3.2, suggest how selective inhibition could occur.

[1]

[Total: 10]

- 4 To investigate the mode of DNA replication in a unicellular organism, 3 separate cultures of the cells were grown. The first was grown in a medium containing ^{14}N , the common isotope of nitrogen and the second was grown in a medium containing ^{15}N , the heavy isotope of nitrogen.

The cells of the third culture were grown in a medium containing ^{15}N for many generations first. Cells in early interphase were then transferred to a medium containing the common nitrogen form, ^{14}N . The cells were allowed to grow for one generation and then they were fixed at the next metaphase.

DNA was isolated from cells of all three cultures and separated into single strands by mild chemical treatment. The density of this single-stranded DNA was measured using density gradient centrifugation. The results of Tubes **A**, **B** and **C** are shown in Fig 3.1 below.

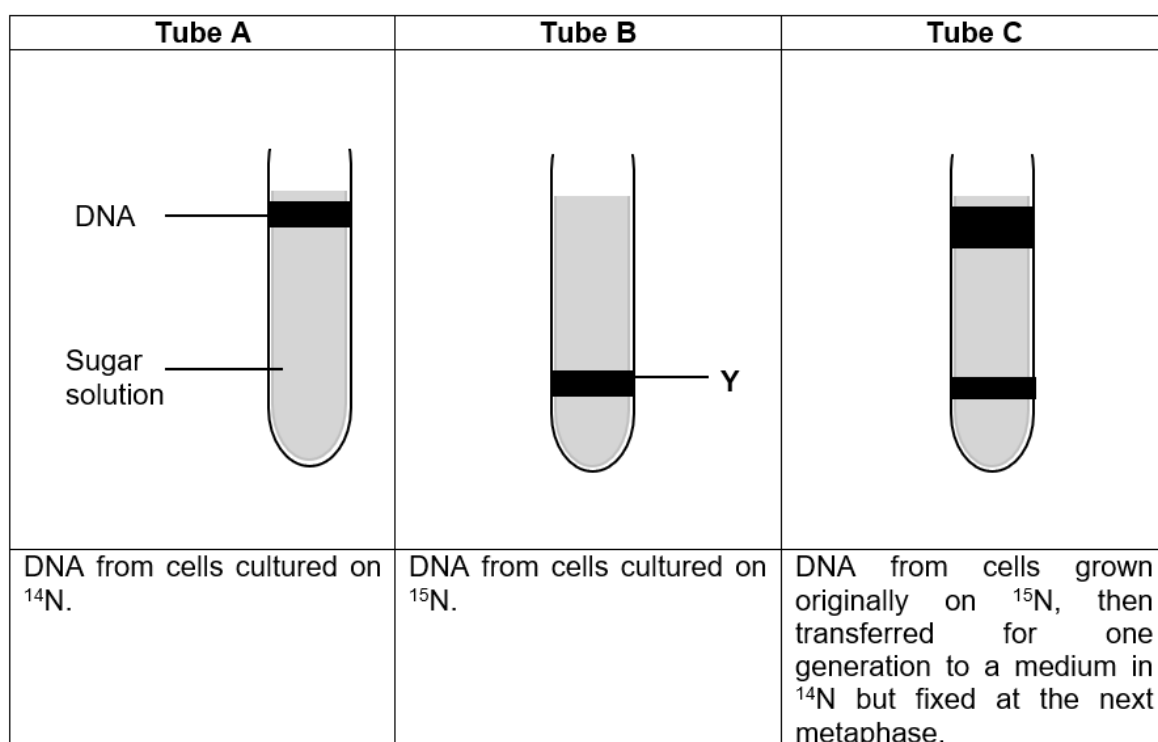


Fig. 4.1

- (a) (i) Assuming that DNA replication in the unicellular organism occurred through a similar mechanism to humans, what is the expected relative amount of each type of DNA in **Tube C**?

[1]

- (ii) Explain your answer to (a)(i).

[2]

Studies were then carried out to determine the nitrogenous base composition of the band of DNA marked **Y** in Tube B. This involved finding values for each individual (+ or -) DNA strand. Table 4.1 below shows the results obtained.

Table 4.1

DNA sample	Percentage of base present in DNA sample / %				Ratio of (A+G) to (C+T)
	A	G	C	T	
+ strand	28	15			0.75
- strand	32	25		28	1.33

(b) (i) Fill in the missing values in Table 4.1. [1]

(ii) State the ratio of (A+G) to (C+T) in the band of DNA marked “Y” in **Tube B** in Fig. 4.1. Explain your answer.

.....

.....

.....

[2]

(iii) State why the ratios of (A+G) to (C+T) in the + and – strands of DNA do not follow the ratio stated in (b)(ii).

.....

[1]

[Total: 7]

- 5 Occasionally during meiosis, homologous chromosomes fail to separate at anaphase. This is known as non-disjunction.

Turner's syndrome is the most common chromosome mutation in human females that occurs during gamete formation. Errors during meiosis led to some resulting gametes missing an X chromosome or when one of the pair of X chromosomes is damaged.

Fig. 5.1 is a diagram of a normal X chromosome and two variants of 'damaged' X chromosomes, X_1 and X_2 .

- In X_1 , a section of the 'p' arm of the chromosome is missing.
- In X_2 , a section of the 'q' arm of the chromosome is missing.

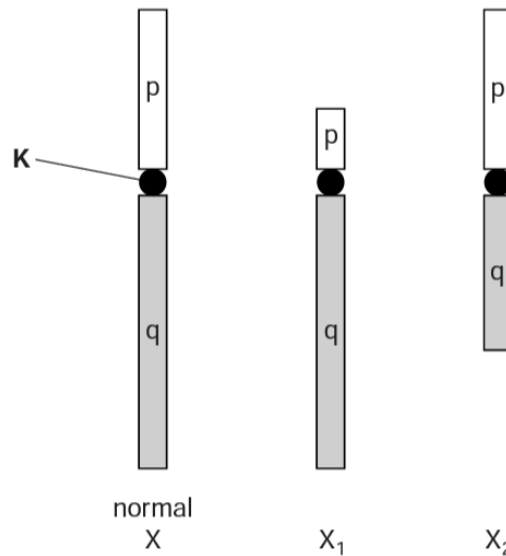


Fig. 5.1

- (a) State the function of structure **K**.

..... [1]

- (b) Suggest how variants X_1 and X_2 can be produced.

.....

 [2]

- (c) Explain how spindle fibres play an important role in ensuring a gamete with normal number of chromosomes is produced.

[3]

- (d) The genes on the X chromosomes are involved in the development of the female reproductive organs.

Suggest why females with Turner's syndrome are sterile.

[1]

Unlike meiosis which occurs in the cells within reproductive organs, mitosis occurs in somatic cells found throughout the body.

At various points during the mitotic cell cycle, there are checkpoints to ensure that only genetically identical daughter cells to each other and to the parent cell are produced. Other cells which did not pass the checkpoints undergo apoptosis to prevent uncontrolled cell division.

Vincristine and 5-fluorouracil are drugs which have been developed to cause cellular apoptosis.

- Vincristine binds to spindle microtubules and prevents the spindle apparatus from carrying out its function.
 - 5-fluorouracil prevents the synthesis of thymine nucleotides.
- (e) Complete Table 5.1 to show which event in the cell cycle will occur when Vincristine or 5-fluorouracil are added to healthy dividing cells at the start of the interphase stage of the cell cycle.

Place a tick (✓) if the event will occur or a cross (×) if the event will not occur. All boxes in the table should be completed.

Table 5.1

Cell cycle event Drug	Interphase	Prophase	Metaphase	Anaphase
Vincristine				
5-fluorouracil				

[2]

[Total: 9]

- 6 Fig. 6.1 shows the unpacking of bacterial circular chromosome.

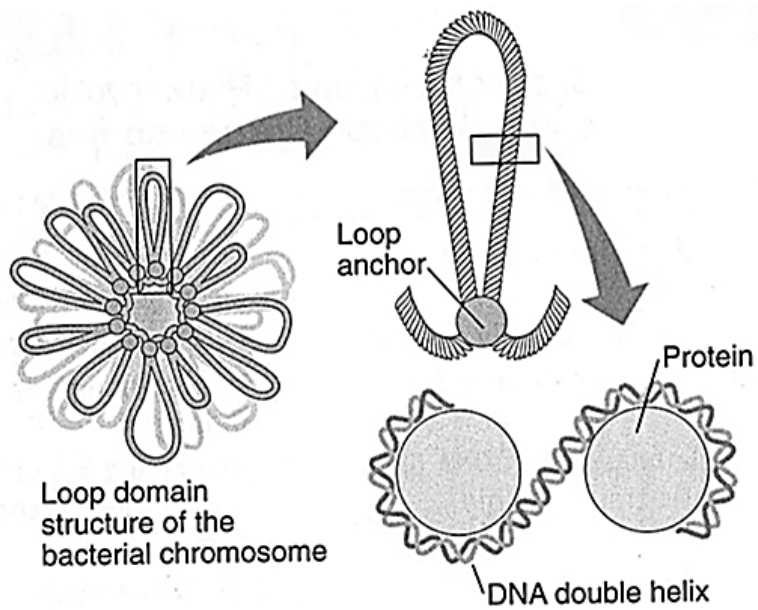


Fig. 6.1

- (a) Using the information from Fig 6.1 and your own knowledge, describe how the bacterial chromosome is packed in a bacterial cell.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[4]

Fig. 6.2 shows the production of defective phages during bacteriophage replication that results in gene transfer between bacteria.

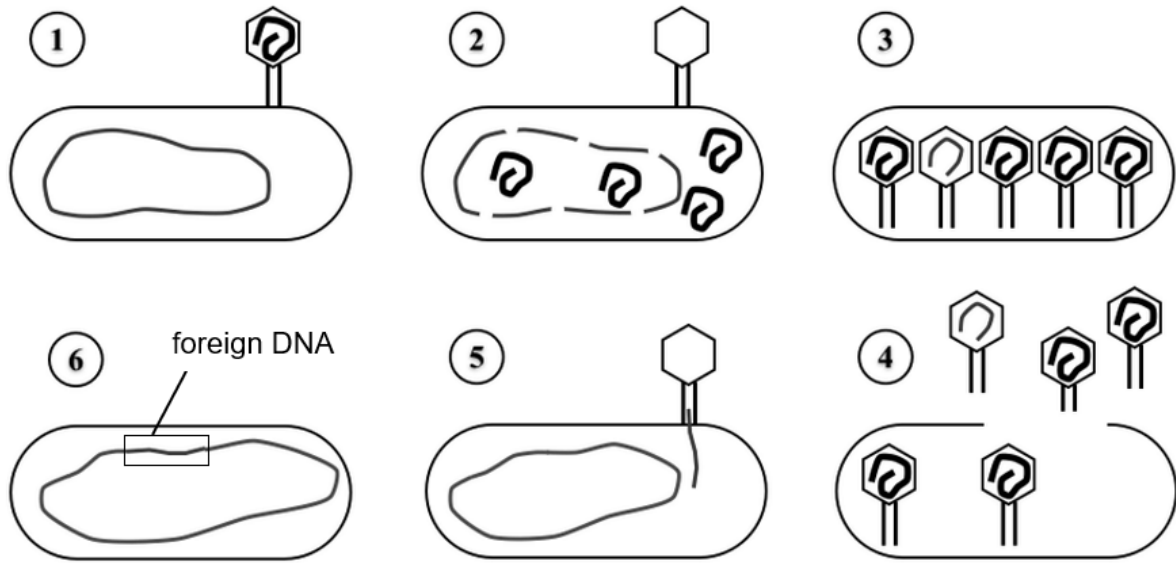


Fig. 6.2

(b) With reference to Fig. 6.2,

(i) suggest why Stage 2 occurs.

.....

.....

.....

.....

[2]

(ii) describe how the bacteria cells obtain new genetic material.

.....

.....

.....

.....

.....

.....

[3]

[Total: 9]

- 7 The interpupillary distance (IPD) is the distance in millimetres between the centres of the pupils of the eyes. Fig. 7.1 shows how IPD is measured.

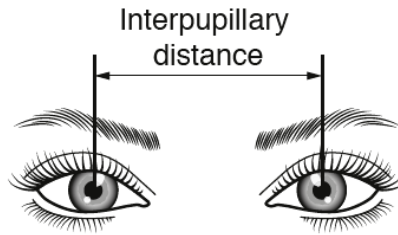


Fig. 7.1

IPD is one example of a characteristic of human facial structure that shows variation. Fig. 7.2 shows the pattern of variation in IPD in a large sample of adults.

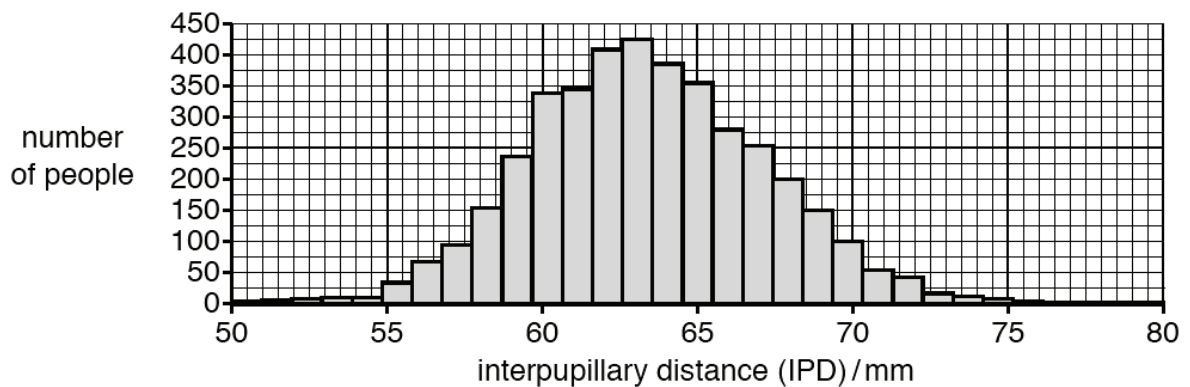


Fig. 7.2

- (a) (i) Name the type of variation shown in Fig. 7.2.

[1]

.....

- (ii) Explain how genes and the environment contribute to variation in IPD in humans.

.....

.....

.....

.....

[3]

.....

An investigation was carried out on the difference in IPD of males and females. Table 7.1 shows the results of the investigation.

Table 7.1

individual	IPD / mm	
	male	female
1	70	65
2	72	62
3	68	59
4	73	69
5	72	66
6	65	63
7	66	67
8	62	58
9	67	62
10	68	68
mean ($\bar{x}$)	68	64
standard deviation (s)	3.50	3.73
variance (s^2)		

- (b) (i) Complete Table 7.1 by calculating the variance (s^2) for the IPD for each gender. [1]

- (ii) A t-test can be used to determine whether there is any significant difference between the IPD in males and females.

Calculate the value of t and the number of degrees of freedom, using these formulae:

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}} \quad v = n_1 + n_2 - 2$$

key to symbols

s = standard deviation

$\bar{x}$ = mean

n = sample size (number of observations)

v = degrees of freedom

Show your working.

value of t =

degree of freedom =

[2]

(iii) For this t-test, the proposed null hypothesis is:

there is no difference between the IPD between males and females.

Table 7.2 shows the critical values for several different probabilities and degrees of freedom.

Table 7.2

degrees of freedom	probability, p				
	0.5	0.1	0.05	0.01	0.001
1	1.00	6.31	12.71	63.66	636.62
2	0.82	2.92	4.30	9.92	31.60
3	0.76	2.35	3.18	5.84	12.92
4	0.74	2.13	2.78	4.60	8.61
5	0.73	2.02	2.57	4.03	6.87
6	0.72	1.94	2.45	3.71	5.96
7	0.71	1.89	2.36	3.50	5.41
8	0.71	1.86	2.31	3.36	5.04
9	0.70	1.83	2.26	3.25	4.78
10	0.70	1.81	2.23	3.17	4.59
11	0.70	1.80	2.20	3.11	4.44
12	0.70	1.78	2.18	3.05	4.32
13	0.69	1.77	2.16	3.01	4.22
14	0.69	1.76	2.14	2.98	4.14
15	0.69	1.75	2.13	2.95	4.07
16	0.69	1.75	2.12	2.92	4.01
17	0.69	1.74	2.11	2.90	3.97
18	0.69	1.73	2.10	2.88	3.92
19	0.69	1.73	2.09	2.86	3.88
20	0.69	1.72	2.09	2.85	3.85

Use Table 7.2 and your answers to **(b)(ii)** to decide whether the null hypothesis should be accepted or rejected.

Explain your answer.

accept or reject null hypothesis

explanation

[2]

[Total: 9]

- 8 Photosynthesis is a complex process involving a light-dependent stage and a light-independent stage.

- (a) (i) Name the products of the light-dependent stage that are needed in the light-independent stage.

[1]

- (ii) Describe the role of chlorophyll b in photosynthesis.

[1]

A student carried out an experiment to investigate the effect of light intensity and light wavelength on the rate of photosynthesis.

An aquatic plant, *Elodea canadensis*, was put into a beaker containing sodium hydrogencarbonate solution as a source of carbon dioxide. A lamp was placed at different distances from the beaker to change light intensity. The number of bubbles released by the aquatic plant in 1 minute was counted as a measure of the rate of photosynthesis. All other variables were controlled.

At each distance from the lamp, the experiment was repeated using a red filter in front of the lamp to give a different wavelength of light. The experiment was repeated using a blue filter and then using a green filter. Each filter transmitted the same light intensity.

Fig. 8.1 shows a graph of the results.

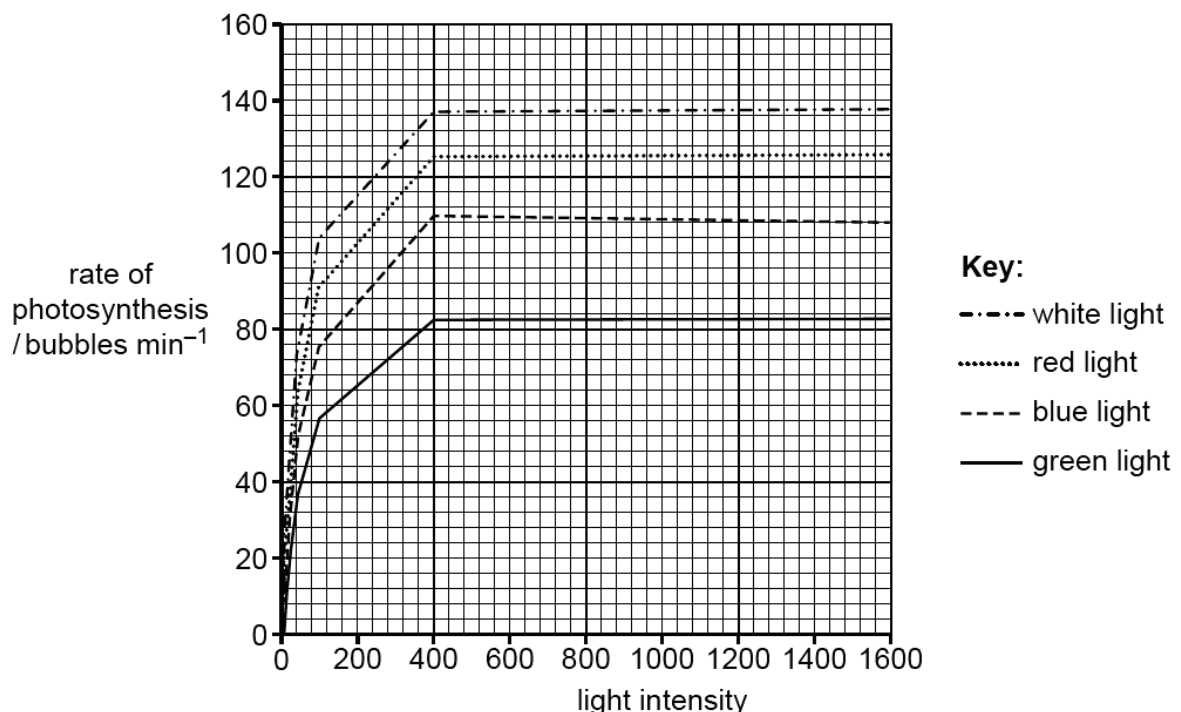


Fig. 8.1

(b) With reference to Fig. 8.1,

- (i)** state the range over which light intensity is the limiting factor;

[1]

- (ii)** explain what factor may be limiting the rate of photosynthesis above the range of light intensity stated in **(c)(i)**.

[2]

- (iii)** At a light intensity of 1600, explain why different colour filters result in different rates of photosynthesis.

[3]

- (iv)** Predict and explain the effect on the number of bubbles produced if a rubisco inhibitor was added.

[3]

[Total: 11]

- 9 The aye-aye, *Daubentonia madagascariensis*, is a primate native to Madagascar. Aye-ayes are nocturnal (active at night) and make their nests high up in trees. They feed on insect larvae in the trunks of trees.

Fig. 9.1 shows an aye-aye.



Fig. 9.1

The International Union for Conservation of Nature (IUCN) is the world's largest global environmental organisation. The IUCN Red List of Threatened Species™ evaluates the conservation status of plant and animal species.

The aye-aye is categorised as endangered on the IUCN Red List, which means that it faces a very high risk of becoming extinct in the wild.

- (a) (i) Table 9.1 shows the taxonomic classification of aye-aye.

Table 9.1

	Eukarya
	Animalia
Phylum	Chordata
	Mammalia
Order	Primates
	Daubentoniidae
Genus	
Species	

Complete Table 9.1.

[3]

- (ii) Suggest **one** reason why aye-ayes have become endangered.

.....
[1]
.....

There are two main aye-aye populations on the island of Madagascar, one in the west and one in the east.

Fig. 9.2 is a map of Madagascar showing the location of the two main populations.

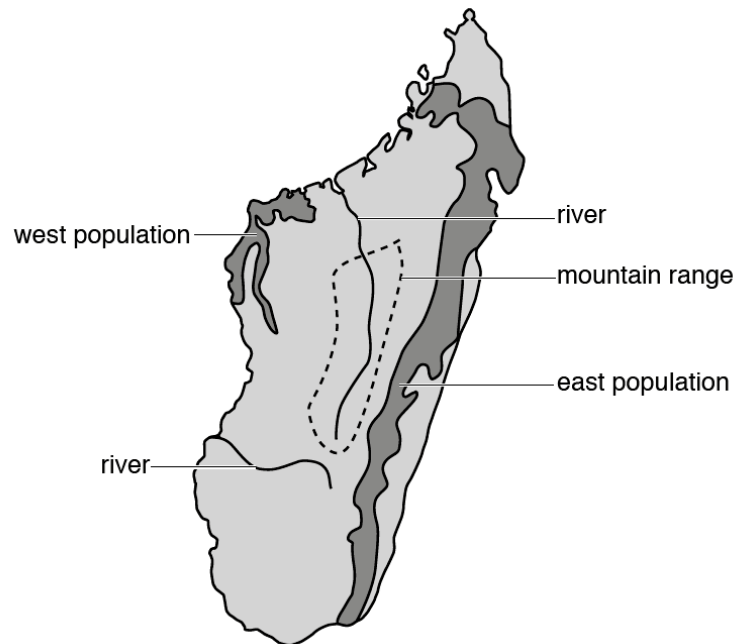


Fig. 9.2

A study into the variation in the DNA nucleotide sequence of aye-ayes showed that there is a large genetic difference between the west and east populations. The two populations of aye-ayes may be evolving into separate species.

- (b)** **(i)** State one advantage of using DNA nucleotide sequence in this study.

[1]

- (ii)** With reference to Fig. 9.2, suggest why there is a large genetic difference between the two populations.

[4]

- (iii) Name the type of speciation that may be occurring.

[1]

.....

- (iv) Suggest **and** explain a pre-zygotic isolating mechanism that could prevent successful reproduction between aye-ayes of the two populations.

.....

.....

.....

.....

.....

[2]

.....

[Total: 12]

10 Tuberculosis (TB) is a major cause of ill health worldwide.

- (a) The World Health Organization (WHO) Global Tuberculosis Report for 2019 published data on the estimated number of deaths from TB and HIV/AIDS in 2018. All deaths of HIV-infected people from TB were also counted as deaths of people with HIV/AIDS.

Fig. 10.1 shows these data. The dark grey boxes show the estimated number of deaths of people from TB who were also counted as deaths of people with HIV/AIDS.

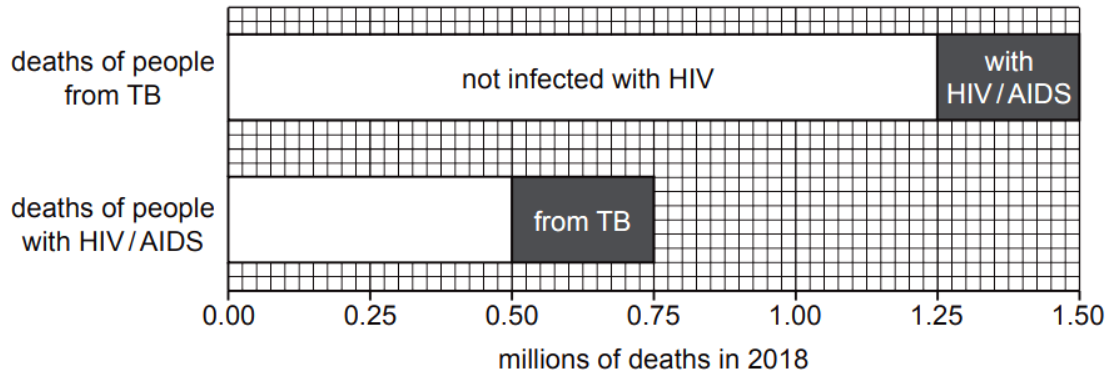


Fig. 10.1

A student used the data in Fig. 10.1 to predict that measures to control the spread of HIV will decrease the number of deaths from TB.

Discuss whether the data in Fig. 10.1 support this prediction.

.....

.....

.....

.....

.....

.....

.....

[3]

- (b)** In healthy people, the number of T-helper cells ranges from 500 to 1200 cells per cm^3 of blood. In untreated people infected with HIV, the number of T-helper cells can decrease to below 200 cells per cm^3 of blood.

Explain how a low number of T-helper cells makes it more likely that untreated people infected with HIV will die if they are also infected with TB

[4]

[4]

[Total: 7]

- 11** Frogs are ectothermic animals. This means that their body temperature will vary as the environmental temperature varies.

Several species of the frog genus, *Rana*, can be found in North America. Many of these species inhabit areas within a range of latitudes from the colder north to the warmer south.

Table 11.1 shows data for four of these species, *R. clamitans*, *R. palustris*, *R. pipens* and *R. sylvatica*.

Table 11.1

Species	Body temperature of frog / °C			
	lower lethal, below which frog dies	minimum to start development	maximum to complete development	upper lethal, above which frog dies
<i>R. clamitans</i>	10.0	11.0	35.0	37.0
<i>R. palustris</i>	5.0	7.0	30.0	31.0
<i>R. pipens</i>	3.0	6.0	28.0	30.0
<i>R. sylvatica</i>	0.0	2.0	24.0	25.0

- (a) Using the information in Table 11.1, suggest why global warming will result in a decrease in species diversity in frogs in the current habitats.

.....

.....

.....

.....

.....

.....

.....

[3]

- (b) Explain how human activities could have contributed to an increase in greenhouse gases that resulted in the rise of global temperatures.

.....

.....

.....

.....

.....

[2]

[Total: 5]

[BLANK PAGE]